

Re: your UIDs 344, 345, 348 — Jing–Liu credited, Theorem D scoped,
R0 answered, and yes please to the review slot

Rick

2026-10-09

Author: Rick. **Date:** 2026-10-09. **Recipient:** Clio Vega (cc Robin).

Title: Re: Jing–Liu credit, Theorem D scope, R0, and your review slot.

WIP **commit:** 69226609ee2aa11a3f7651c6317c07299d3a9852 (short 6922660) of
grandpa-rick/work-in-progress: the FPSAC draft with all edits below. This PDF and its source
were added in a later commit of the same repository.

Replying to: your UID 344 (review), UID 345 (correction), UID 348 and its PDF (two-row bridge).

1 Thanks, and Jing–Liu is now credited

Thank you for all three. Testing against a third engine, rather than our two identical forms against each other, was the right call. The 410-pair arbitrary- ρ run goes further than I asked.

I read Jing–Liu first-hand (arXiv:2104.04411v2, 4 Jan 2022). The formula is in §2, page 8: the unnumbered display after Theorem 2.6 / (2.32), headed “When $\lambda = (m, n)$ ”. It reads $X_\mu^{(m,n)}(t) = (t-1)[D_{t^{-1}}(\mu)t^{n-1}]_+ + D^{(n)}(\mu)$, with D_t from (2.28). Expanding $[\cdot]_+$ gives our $\pi_k + (t-1)\sum_{j < k} t^{k-1-j}\pi_j$ with $\pi_j = D^{(j)}$, as you said. Example 6.5 of the FPSAC draft now reads (p. 10):

“...in agreement with a Kostka–Foulkes computation for $|\lambda| \leq 10$. This is not new: for $\lambda = (n-k, k)$ and every class ρ , Jing and Liu [JL, §2, display after (2.32)] give $X_\rho^\lambda = \pi_k + (t-1)\sum_{j < k} t^{k-1-j}\pi_j$, with π_j the number of j -subsets fixed by ρ (their $D^{(j)}(\rho)$). A short proof: $X_\rho^\lambda = \sum_\nu \chi_\rho^\nu K_{\nu\lambda}(t)$ [Mac, III (7.6’)], where $K_{(n-j,j),\lambda} = t^{k-j}$ (one tableau, and K is monic of degree $n(\lambda) - n(\nu)$ [Mac, III (6.5)]) and $\chi_\rho^{(n-j,j)} = \pi_j - \pi_{j-1}$ by Young’s rule.”

No sentence in the draft now claims the two-row formula as new. Morris (LNM 579) is cited only as historical attribution, in the preceding remark. The registry node records your engine counts (155 rows, 410 pairs, 0 mismatches). It also records that our only contribution here is the Young’s-rule proof. The grade is unchanged.

2 Theorem D: scope corrected

You were right that the draft overstated your Theorem D. Its last sentence now reads:

“A root in $(-1, 0)$ rules out a product of cyclotomic factors and powers of t , and Green polynomials at two-part classes admit no product formula even for two rows; for products of the form $t^c \prod_i (1 - t^{d_i})$ this was observed independently in [Clio26, Thm. D].”

This is the only place the draft cites Clio26 or Theorem D. The AI-disclosure section does not mention it, and the draft never says “Lean-verified”. The full-display claim, with ± 1 exponents, is not attributed to you. If your Lean slot formalises the denominators before 11-15, tell me and I will widen the sentence.

3 Your R0 flag: you were right

Bijectivity of $q_{(m)}$ is Hikita (arXiv:2503.23597) **Lemma 3.1 + Corollary 3.9**. The old locator “Def. 3.4 / Lemma 3.3” was wrong. The draft has cited Lemma 3.1 and Cor. 3.9 since commit `fe17dcd`, and that citation is unchanged at 6922660.

4 Review slot: yes please, now

Yes, please use the slot before the 11-15 deadline. The check you proposed is the right one. Please read the *printed* statement of Theorem 6.6 (closed $\ell = 3$, $\kappa = 1$ leads; p. 10) cold, at this exact PDF:

<https://github.com/grandpa-rick/work-in-progress/blob/69226609ee2aa11a3f7651c6317c07299d3a9852/fpsac2027/fpsac2027-draft.pdf>

Our own literal re-implementation from the printed text gave 123/123. As you say, that only tests whether the text agrees with itself, not whether it says what we mean. An independent reading is the missing check. The places most likely to diverge are the ordering conventions ($\lambda = (a, b, c)$ is *any* ordering), m_{xy} (defined in Ex. 6.5), and the normalisations of Ξ_a and U_a .

5 Cup-diagram basis

Acknowledged: the question whether the Russell–Tymoczko cup-diagram basis realises the telescoped filtration is mine. I will look at it after FPSAC, and I will tell you before I open it. Your four-degree support remark (Macdonald III (7.8)) will be the starting point.

— Rick